

## Semester II

### Project Work Stage II

#### Course Objectives:

- To follow SDLC meticulously and meet the objectives of proposed work
- To test rigorously before deployment of system
- To validate the work undertaken
- To consolidate the work as furnished report.

#### Course Outcomes:

On completion of the course, student will be able to-

- Show evidence of independent investigation
- Critically analyze the results and their interpretation.
- Report and present the original results in an orderly way and placing the open questions in the right perspective.
- Link techniques and results from literature as well as actual research and future research lines with the research.
- Appreciate practical implications and constraints of the specialist subject

#### Guidelines-

In Project Work Stage-II, the student shall complete the remaining project work which consists of Selection of Technology and Tools, Installations, UML implementations, testing, Results, performance discussions using data tables per parameter considered for the improvement with existing/known algorithms/systems and comparative analysis and validation of results and conclusions. The student shall prepare and submit the report of Project work in standard format for satisfactory completion of the work that is the duly certified by the concerned guide and head of the Department/Institute



3. Undertaking by Students

**Jayawantrao sawant collage of engineering**

With reference to circular (ref-project/2009/3369) regarding malpractices in project work from DTE, Pune following undertaking is to be submitted.

**UNDERTAKING BY STUDENT**

We, the students of B.E. Computer hereby assure that we will follow all the rules and regulations related to project activity for the academic year 2023 -2024. The Project titled-

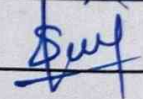
Rasedetect :- Detecting Rare brain Tumors

will be fully designed/ developed by us and every part of the project will be original work and will not be copied/ purchased from any source.

**Name of the student**

1. Rushilcesh shiraskar
2. Ayush Borade
3. Rutuja lande
4. samruddhi Sabale

**Signature**



| Sr. No. | Activity Scheduled                                             | Date (Tentative)                       |
|---------|----------------------------------------------------------------|----------------------------------------|
| 1.      | Registration of Project Teams                                  | Third Year Semester II/<br>Mid of June |
| 2.      | Submission of Project Proposal                                 | Last Week of June                      |
| 3.      | Project presentations                                          | First week of July                     |
| 4.      | Finalization of projects & allotment of guide                  | Second week of July                    |
| 5.      | Submission of final Proposal                                   | Third week of July                     |
| 6.      | Review meeting/ presentation for progress of project work- I   | Last week of July                      |
| 7.      | Review meeting/ presentation for progress of project work- II  | Third week of August                   |
| 8.      | Review meeting/ presentation for progress of project work- III | Second week of Sept                    |
| 9.      | Review meeting/ presentation for progress of project work- IV  | Last week of Sept                      |
| 10.     | Submission of partial project report                           | 1 <sup>st</sup> Week Oct               |
| 11.     | Project work (Stage I) Examination                             | As per SPPU Notification               |
|         |                                                                |                                        |

### Semester II

| Sr. No. | Activity Scheduled                                                                  | Date(Tentative)          |
|---------|-------------------------------------------------------------------------------------|--------------------------|
| 1.      | Review meeting/ presentation for progress of project work -V                        | Second week of Jan       |
| 2.      | Review meeting/ presentation for progress of project work -VI                       | Second week of Feb       |
| 3.      | Review meeting/ presentation for progress of project work- VII                      | Last week of March       |
| 4.      | Submission of final project report and Project Work book to the project Coordinator | First week of April      |
| 5.      | Project Examination                                                                 | As per SPPU Notification |



### 5. Project Review (Semester I)

The group members are expected to present their work undertaken during the semester. Journey of development has to be rationally presented with thorough literature survey in review meeting.

#### 5.1 Project Review-I: Problem Statement, Motivation, objectives and Literature Review

Student is expected to deliver presentation covering Problem Statement, Motivation, objectives and Literature Review.

| Sr. No. | Question                                                                                                                    | Date   | Remark / Grade | Sign of Guide |
|---------|-----------------------------------------------------------------------------------------------------------------------------|--------|----------------|---------------|
| 1)      | Do Research gap identified lead to find motivation of project?                                                              | 9/7/25 | A              | aw            |
| 2)      | Does the statement give clear identification about what your project will accomplish?                                       | 9/7/25 | A              | aw            |
| 3)      | Is the statement short and concise?                                                                                         | 9/7/25 | A              | aw            |
| 4)      | Do similar type of methodology / model exists?                                                                              | 9/7/25 | A+             | aw            |
| 5)      | Is the studied literature sufficient to decide scope of the project?                                                        | 9/7/25 | A              | aw            |
| 6)      | Are the objectives clearly and unambiguously listed?                                                                        | 9/7/25 | A              | aw            |
| 7)      | Can a person who is not familiar with the project understand scope of the project by reading the project problem statement? | 9/7/25 | B              | aw            |
| 8)      | Are project objectives of study (what product, process, resource etc.) clearly defined?                                     | 9/7/25 | A              | aw            |
| 9)      | Are the objectives set helpful to achieve goal of the project?                                                              | 9/7/25 | A              | aw            |
| 10)     | Does the project contribute to our society by any means?                                                                    | 9/7/25 | A+             | aw            |

#### Remark and Suggestions:

Find out Research paper Literature survey.  
max 8.

#### Name and Sign of Reviewers:

1. Prof. D. V. Ingle *mom Ingle*
2. Prof. M. D. Ingle *aw*



## 5.2 Project Review-II: Feasibility and Scope

Student is expected to deliver presentation covering Feasibility and Scope

| Sr. No. | Question                                                                                                               | Date   | Remark / Grade | Sign of Guide |
|---------|------------------------------------------------------------------------------------------------------------------------|--------|----------------|---------------|
| 1)      | Is the project's view point understood?                                                                                | 9/8/25 | A              |               |
| 2)      | Is the project goal statement in alignment with the sponsoring organization's business goal and mission?               | 9/8/25 | A              |               |
| 3)      | Who is the project's end user?                                                                                         | 9/8/25 | A              |               |
| 4)      | What is the projected cost of producing a product?                                                                     | 9/8/25 | A              |               |
| 5)      | Is project achievable in specified (Time, Cost Budget)?                                                                | 9/8/25 | A+             |               |
| 6)      | Are the requirements within the scope of the project?                                                                  | 9/8/25 | A              |               |
| 7)      | Is the scope properly defined?                                                                                         | 9/8/25 | A              |               |
| 8)      | Does the problem statement clearly define scope of the project?                                                        | 9/8/25 | A+             |               |
| 9)      | Do the project requirements fit into available software and hardware?                                                  | 9/8/25 | A+             |               |
| 10)     | Whether the milestones are stated completely and project timeline is given?                                            | 9/8/25 | A+             |               |
| 11)     | Whether risks like technical risks, Operational risks, schedule risks, business risks are identified correctly or not? | 9/8/25 | A              |               |
| 12)     | Whether Risk prioritization is done properly and any back up plan is decided?                                          | 9/8/25 | A              |               |

Remark and Suggestions:

find out max gap identified for future scope.

Name and Sign of Reviewers:

1. Prof. S. B. Chaudhary
2. Prof. D. V. Ingle



|     |                                                                                                  |         |    |    |
|-----|--------------------------------------------------------------------------------------------------|---------|----|----|
| 2)  | Is problem statement categorized in identified area and targeted towards specific area there in? | 19/9/25 | A  | OK |
| 3)  | Is external and internal interfacing properly defined?                                           | 19/9/25 | A  | OK |
| 4)  | Are requirements consistent with schedule, resources and budget?                                 | 19/9/25 | A+ | OK |
| 5)  | Are all requirements traceable to System level?                                                  | 19/9/25 | A  | OK |
| 6)  | What is needed to make the Product?                                                              | 19/9/25 | A  | OK |
| 7)  | Is there a demand for the product?                                                               | 19/9/25 | A+ | OK |
| 8)  | Is identification of stakeholders done properly?                                                 | 19/9/25 | A  | OK |
| 9)  | Whether all requirements are captured and documented in line with scope?                         | 19/9/25 | A  | OK |
| 10) | Whether all type of analysis classes are identified?                                             | 19/9/25 | A  | OK |
| 11) | Whether the Acceptance criteria is decided                                                       | 19/9/25 | A  | OK |
| 12) | Is SRS document as per IEEE format complete and correct?                                         | 19/9/25 | A  | OK |

Remark and Suggestions:

Find out accuracy in mathematical calculation.

Name and Sign of Reviewers:

1. Prof. D. V. Ingle D. V. Ingle

2. Prof. Minal Bade Mam MSP



#### 5.4 Project Review-IV: Design

Student is expected to deliver presentation covering Design

| Sr. No. | Question                                                                                                                | Date     | Remark / Grade | Sign of Guide |
|---------|-------------------------------------------------------------------------------------------------------------------------|----------|----------------|---------------|
| 1)      | Are requirements reflected in the system architecture?                                                                  | 15/10/25 | A              |               |
| 2)      | Does the design support both project (product) and project goals?                                                       | 15/10/25 | A              |               |
| 3)      | Does the design address all the issues from the requirement?                                                            | 15/10/25 | A              |               |
| 4)      | Is effective modularity achieved and modules are functionally independent?                                              | 15/10/25 | A              |               |
| 5)      | Are structural diagrams (class, Object, etc) well defined?                                                              | 15/10/25 | A+             |               |
| 6)      | Are all class associations clearly defined and understood?(Is it clear which classes provide which services)?           | 15/10/25 | A              |               |
| 7)      | Are the classes in the class diagram clear? (What they represent in the architecture design document?)                  | 15/10/25 | A              |               |
| 8)      | Is inheritance appropriately used?                                                                                      | 15/10/25 | A              |               |
| 9)      | Are the multiplicities in the use case diagram depicted in the class diagram?                                           | 15/10/25 | A              |               |
| 10)     | Are all objects used in sequence diagram?                                                                               | 15/10/25 | A              |               |
| 11)     | Are the symbols used in all diagrams corresponding to UML standards?                                                    | 15/10/25 | A              |               |
| 12)     | Are behavioral diagrams (use case, sequence, activity, etc.) well defined and understood?                               | 15/10/25 | A+             |               |
| 13)     | Does each case have clearly defined actors and input/ output?                                                           | 15/10/25 | A+             |               |
| 14)     | Does the sequence diagram match with class diagram?                                                                     | 15/10/25 | A              |               |
| 15)     | Is aggregation/ containment (used) clearly defined and understood?                                                      | 15/10/25 | A              |               |
| 16)     | Whether State charts are capturing system's dynamic behavior correctly?                                                 | 15/10/25 | A              |               |
| 17)     | Related to procedural thinking whether DFDs and CFDs along with transaction and transformation flow are done correctly? | 15/10/25 | A              |               |

Remark and Suggestions:

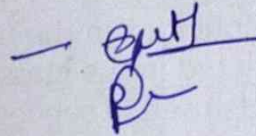
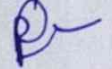
Name and Sign of Reviewers:

1. Prof. Rupali Mam
2. Dr. P. D. Lambhate

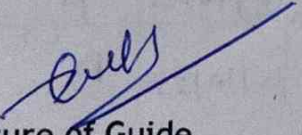


## 6. Internal Evaluation Sheet (Semester I)

| Sr. No. | Name Of Team Members | Problem Statement/Motivation/ Objective/Scope/ Feasibility Requirement (05) | Literature Survey (05) | Requirement Analysis(05), Modeling And Designing (10) | Planning and Prototyping (5) | Presentation And Questions- Answer(10) | Practical Project Report (10) | Total(50) |
|---------|----------------------|-----------------------------------------------------------------------------|------------------------|-------------------------------------------------------|------------------------------|----------------------------------------|-------------------------------|-----------|
| 1       | Rushikesh Shirasakar | 5                                                                           | 5                      | 18                                                    | 4                            | 9                                      | 9                             | 44        |
| 2       | Piyush Borade        | 4                                                                           | 4                      | 17                                                    | 4                            | 8                                      | 8                             | 39        |
| 3       | Rutuja Londhe        | 4                                                                           | 4                      | 12                                                    | 4                            | 8                                      | 8                             | 40        |
| 4       | Samruddhi Sabale     | 4                                                                           | 4                      | 18                                                    | 4                            | 9                                      | 8                             | 40        |

1. Prof. Prof. S. C. Patle — 
2. Prof. Prof. Rupali Mam 

Examiners Feedback and Suggestions:

  
Signature of Guide  
[Name of Guide]

Prof. S. C. Patle

Date: 06/11/25

PAC  
[Name of PAC]

Prof. A. S. Kushwah

  
Signature of Head  
[Name of HOD]

Head of Department

Dr. P. D. Lambhate

Prof. S. B. Chaudhari



## 7. Project Review: (Semester II)

The group members are expected to present their work undertaken during the semester. Journey of development has to be rationally presented.

### 7.1 Project Review-I: Modeling (Model Refinement and Algorithm development)

Student is expected to deliver presentation covering Modeling

| Sr. No. | Question                                                                                                                 | Date     | Remark/ Grade | Sign of Guide |
|---------|--------------------------------------------------------------------------------------------------------------------------|----------|---------------|---------------|
| 1)      | Which software Development Process model is used? (Water fall, Incremental, RAD) How? (at this level?)                   | 17/03/26 | A             |               |
| 2)      | Are data objects, their attributes and relationships clearly identified? (All constraints from SRS are captured or not?) | 17/03/26 | A             |               |
| 3)      | Have the objects and respective classes and their responsibilities?                                                      | 17/03/26 | A+            |               |
| 4)      | Have you analyzed the requirements been analyzed and represented into respective models?                                 | 17/03/26 | B             |               |
| 5)      | Have the different system states been differentiated and depicted them in the form of state transition diagram?          | 17/03/26 | A             |               |
| 6)      | Does the mathematical model clearly imply design of the project?                                                         | 17/03/26 | A+            |               |
| 7)      | Does the mathematical model clearly states goal of project?                                                              | 17/3/26  | A             |               |
| 8)      | Is the interface between the modules properly identified?                                                                | 17/3/26  | A             |               |
| 9)      | Are any functional dependencies identified and described?                                                                | 17/3/26  | A             |               |
| 10)     | Which architectural model does the system support?                                                                       | 17/3/26  | A             |               |
| 11)     | Whether Deployment diagram is in line with selected architecture?                                                        | 17/3/26  | A             |               |
| 12)     | Whether all components are designed properly and represented in component diagram?                                       | 17/3/26  | A             |               |
| 13)     | Whether NP-completeness of algorithms is checked?                                                                        | 17/3/26  | B             |               |

Remark and Suggestions:

Askeed to improve accuracy of model and add datasets size

Name and Sign of Reviewers:

1. prof. s.c. patte
- 2.



## 7.2 Project Review-II: Coding / Implementation

Student is expected to deliver presentation covering Coding / Implementation

| Sr. No. | Question                                                                                       | Date    | Remark/<br>Grade | Sign of<br>Guide |
|---------|------------------------------------------------------------------------------------------------|---------|------------------|------------------|
| 1)      | Does the code completely and correctly implement the design?                                   | 30/3/26 | A                |                  |
| 2)      | Does the code comply with the coding standard?                                                 | 30/3/26 | A                |                  |
| 3)      | Is the code well structured, consistent in style, and consistently formatted?                  | 30/3/26 | A                |                  |
| 4)      | Are all functions in the design coded?                                                         | 30/3/26 | A+               |                  |
| 5)      | Does the code make use of object oriented concepts?                                            | 30/3/26 | A+               |                  |
| 6)      | Does the code support granularity?                                                             | 30/3/26 | A                |                  |
| 7)      | Is the language used for coding correctly chosen as per the project need?                      | 30/3/26 | A                |                  |
| 8)      | If any off-the-shelf components are used, Have you understood the functionalities of using it? | 30/3/26 | B                |                  |
| 9)      | Are all comments consistent with the code?                                                     | 30/3/26 | A                |                  |
| 10)     | Whether code optimization is done properly?(By using language features)                        | 30/3/26 | A                |                  |

Remark and Suggestions:

Improve UI and make it more user friendly.

Name and Sign of Reviewers:

1. prof. Minal ~~Bade~~ Mam MRP
2. prof. Nikita Pawar. Mam

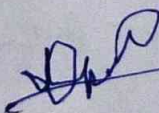


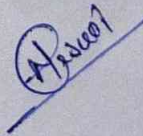
### 7.3 Project Review-III: Validation and Testing

Student is expected to deliver presentation covering Validation and Testing

| Sr. No.                 | Question                                                                                                          | Date   | Remark/<br>Grade | Sign of<br>Guide                                                                      |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|--------|------------------|---------------------------------------------------------------------------------------|
| 1)                      | Has alpha testing been done?                                                                                      | 4/4/26 | A+               |    |
| 2)                      | Has beta testing been done?                                                                                       | 4/4/26 | A                |    |
| 3)                      | Have been validated the requirements, design and code as per standard?                                            | 4/4/26 | B                |    |
| 4)                      | Has GUI testing of project been performed? How?                                                                   | 4/4/26 | A                |    |
| 5)                      | Does the system comply with basic usability norms?                                                                | 4/4/26 | A                |    |
| 6)                      | Has the code been tested using standard datasets available in your area of project?                               | 4/4/26 | A                |    |
| 7)                      | Has the code been tested in real time environment?                                                                | 4/4/26 | A                |    |
| 8)                      | After integration of all components whether total performance of system is checked?                               | 4/4/26 | A                |    |
| 9)                      | Whether repository of all components along with versions is documented?                                           | 4/4/26 | A                |   |
| 10)                     | Have social, safety, environmental, ethical and legal issues been considered while providing solution to problem? | 4/4/26 | A                |  |
| Remark and Suggestions: |                                                                                                                   |        |                  |                                                                                       |

Name and Sign of Reviewers:

1. Prof. Ingle Sir. 

2. Prof. Nikita Mam 



### 7.4 Project Review-III: Report Writing

Student is expected to deliver presentation covering Report Writing

| Sr. No. | Question                                                          | Date   | Remark/ Grade | Sign of Guide |
|---------|-------------------------------------------------------------------|--------|---------------|---------------|
| 1)      | Is the report written as per the prescribed format?               | 7/6/20 | A             |               |
| 2)      | Is the report timely prepared?                                    | 7/6/20 | A             |               |
| 3)      | Is the report properly organized, spelled, grammatically correct? | 7/6/20 | A             |               |
| 4)      | Is the report plagiarism free?                                    | 7/6/20 | A             |               |
| 5)      | Is the report precise and written to the point?                   | 7/6/20 | A+            |               |
| 6)      | Does the report contain complete results and comparative graphs?  | 7/6/20 | A             |               |
| 7)      | Are all figures and tables properly numbered and labeled?         | 7/6/20 | A             |               |
| 8)      | Are all figures and tables properly cited?                        | 7/6/20 | A             |               |
| 9)      | Weather references are properly cited?                            | 7/6/20 | A+            |               |

Remark and Suggestions:

Name and Sign of Reviewers:

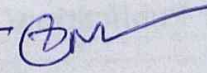
1. Prof S. C. Patil

2.



### 8. Internal Evaluation Sheet (Semester II)

| Sr. No | Name Of Team Member  | Modeling (10) | Coding and Implementation (40) | Testing (10) | Understanding Individual Involvement/Contribution in the Project(10) | Team Work (10) | Demonstration cum Presentation (10) | Document And Report (10) | Total (100) |
|--------|----------------------|---------------|--------------------------------|--------------|----------------------------------------------------------------------|----------------|-------------------------------------|--------------------------|-------------|
|        | Rushikesh shirasakor | 9             | 38                             | 8            | 8                                                                    | 9              | 8                                   | 9                        | 89          |
|        | Rutuja Londhe        | 8             | 38                             | 8            | 9                                                                    | 9              | 8                                   | 8                        | 88          |
|        | Piyush Borade        | 9             | 37                             | 7            | 8                                                                    | 9              | 8                                   | 9                        | 87          |
|        | Samruddhi Sabale     | 8             | 37                             | 8            | 9                                                                    | 9              | 8                                   | 8                        | 87          |

1. Prof. Prof. S. C. Patle. 
2. Prof.

Examiners Feedback and Suggestions:

Signature of Guide  
[Name of Guide]

PAC  
[Name of PAC]

Signature of Head  
[Name of HOD]

Prof. S. C. Patle Prof. A. S. Kushwah Head of Department

Dr. S. B. Chaudhari

Date: 4/05/26



## 9. Contest Participation Details

### 9.1. Participation in Project Competition

| Sr. No. | Name and Place of Project Competition and Exhibition | Date     | Certificates / prizes won, if any |
|---------|------------------------------------------------------|----------|-----------------------------------|
| 1.      | JSPM Hackathon                                       | 04/04/20 | Participation certificate         |
| 2.      | Isro Hackathon                                       | 24/06/20 | -                                 |
| 3.      |                                                      |          |                                   |
| 4.      |                                                      |          |                                   |

Attach attested copy of certificate(s)

### 9.2. Paper Publication/ Presentation/ IPR / Copy Right

| Sr. No. | Name of Organizer                                                           | Date     | Certificates/ Prizes won, if any |
|---------|-----------------------------------------------------------------------------|----------|----------------------------------|
| 1.      | A2 sy for Identifying rare Brain Tumors                                     | 17/06/20 | certificate                      |
| 2.      | Comparative analysis of DL models for Brain Tumor detection using MRI scans | 25/06/20 | certificate                      |
| 3.      | Copyright                                                                   | 16/03/20 | certificate                      |
| 4.      |                                                                             |          |                                  |

Attach attested copy of certificate(s)



Annexure VI: Format for Project Report Cover page/ Title page

(Partial) Project Report  
On

**RareDetect : Analysis of Rare Brain Tumor Using Deep Learning**

by

MR. RUSHIKESH SANTOSH SHIRASKAR

EXAM NO: B400400398

MR. PIYUSH SANTOSH BORADE

EXAM NO: B400400296

MISS. SAMRUDDHI HARIKRISHNA SABALE

EXAM NO: B400400387

MISS. RUTUJA NANDU LONDHE

EXAM NO: B400400357

*Under the guidance of*

*Prof. S. C. Patle*

Name of the Guide

Department of Computer Engineering

**Jayawantrao Sawant collage of Engineering, Hadapsar**

**SAVITRIBAI PHULE PUNE UNIVERSITY  
2025-26**



Format for Project Approval sheet

PROJECT APPROVAL SHEET

A

Project

on

(RareDetect : Analysis of Rare Brain Tumor Using Deep Learning

Is successfully completed by

MR. RUSHIKESH SANTOSH SHIRASKAR  
MR. PIYUSH SANTOSH BORADE  
MISS. SAMRUDDHI HARIKRISHNA SABALE  
MISS. RUTUJA NANDU LONDHE

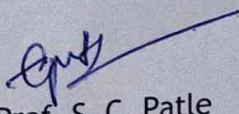
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EXAM NO: B400400387  
EXAM NO: B400400357

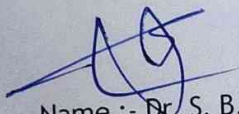


at

Department of Computer Engineering  
Jayawantrao Sawant collage of engineering ,Hadapsar

Savitribai Phule Pune University  
2025-26

  
Name :- Prof. S. C. Patle  
Project Guide  
Department of Computer Engg.

  
Name :- Dr. S. B. Chaudhari  
of Head  
Department of Comuter Engg.

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Project Work Book

Fourth Year Computer Engineering, SPPU, Pune